NOT MEASUREMENT SENSITIVE

MIL-DTL-45150L
30 September 1998
SUPERSEDING
MIL-PRF-45150K
6 April 1997

DETAIL SPECIFICATION

CHASSIS, TRAILER, 2-WHEEL CART TYPE 1/4 TO 3-1/2 TON

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers 2-wheel trailer chassis intended for mounting of various type bodies and equipment used by the United States Armed Forces (see 6.1).

1.2 Classification. Chassis should be of the following models as specified (see 6.2):

Model	Description
M569A1	1/4 Ton
M569A2	3/4 Ton
M116A2E1	3/4 Ton
M116A3	1 Ton
M514	1 Ton
XM545	1 Ton
M103A3	1-1/2 Tons
M103A4	1-1/2 Tons
M492A1	1-1/2 Tons
XM580	1-1/2 Tons
M390	2 Tons

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Tank-automotive and Armaments Command, ATTN: AMSTA-TR-E/BLUE, Warren, MI 48397-5000, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

AMSC N/A FSC 2330

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Model - Continued.	Description - Continued.
M390C	2 Tons
M200A1	2-1/2 Tons
M537	2-1/2 Tons
M353	3-1/2 Tons
M536	3-1/2 Tons

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirement documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplements thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

DEPARTMENT OF DEFENSE

MIL-PRF-3150	- Lubricating Oil, Preservative, Medium.
MIL-PRF-6083	- Hydraulic Fluid, Petroleum Base, for Preservation and Operation.
MIL-PRF-10924	- Grease, Automotive and Artillery
MIL-G-23827	- Grease, Aircraft and Instrument, Gear and Actuator Screw.
MIL-B-46176	- Brake Fluid, Silicone, Automotive, All Weather, Operational and Preservative.

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

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DEPARTMENT OF TRANSPORTATION (DoT)

Federal Motor Carrier Safety Regulations (FMCSR).
Federal Motor Vehicle Safety Standards (FMVSS).

(Application for copies should be addressed to the U.S. Department of Transportation, Superintendent of Documents, Government Printing Office, Washington, DC 20402.)

ENVIRONMENTAL PROTECTION AGENCY (EPA)

Control of Air Pollution from New Motor Vehicles and New Motor Vehicle Engines.

(Application for copies of ERA publications should reference the Code of Federal Regulations, 40 CFR, and the Federal Register and should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

NUCLEAR REGULATORY COMMISSION (NRC)

Code of Federal Regulations (CFR) - Title 10, Parts 30 and 40.

(Copies of the Code of Federal Regulations (CFR) are available from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

DRAWINGS

ARMY

8358991	- Chassis, Trailer, 1-1/2 Tons, 2-Wheel, M103A3.
8358992	- Chassis, Trailer, 1-1/2 Tons, 2-Wheel, M103A4.
8358998	- Chassis, Trailer, 2-1/2 Tons, 2-Wheel, M200A1.
8736243	- Chassis, Trailer, 1 Ton, 2-Wheel, M514.
8736268	- Chassis, Trailer, 3-1/2 Tons, 2-Wheel, M353.
8736315	- Chassis, Trailer, 2 Tons, 2-Wheel, M390C.
8736321	- Chassis, Trailer, 3-1/2 Tons, 2-Wheel, M536.
8736322	- Chassis, Trailer, 2-1/2 Tons, 2-Wheel, M537.
8736324	- Chassis, Trailer, 2 Tons, 2-Wheel, M390.
8736332	- Chassis, Trailer, 1 Ton, 2-Wheel, XM545.
8736417	- Chassis, Trailer, 1-1/2 Tons, 2-Wheel, XM580.
8736829	- Chassis, Trailer, 1-1/5 Tons, 2-Wheel, M492A1.
8736996	- Chassis, Trailer, 1/4 Ton, 2-Wheel, M569A1.
8750054	- Chassis, Trailer, 3/4 Ton, 2-Wheel, M116A2.
8750240	- Chassis, Trailer, 3/4 Tons, 2-Wheel, M116A2E1.
8750253	- Chassis, Trailer, 1 Ton, 2-Wheel, M116A3.

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(Copies of these drawings are available from the U.S. Army Tank-automotive and Armaments Command, ATTN: AMSTA-TR-E/BLUE, Warren, MI 48397-5000.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DoDISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D522 - Standard Test Methods For Mandrel Bend Test of Attached Organic Coatings (DoD Adopted).

(Application for copies of ASTM publications may be obtained from the American Society for Testing and Materials, 100 Barr Harbor Drive, West Conshohocken, PA 19428-2959.)

GENERAL MOTORS CORPORATION

GM 9540P - Accelerated Corrosion Test.

(Copies of GM publications may be obtained from General Motors Corporation, c/o Global Engineering, 15 Inverness Way, Englewood, CO 80112.)

2.4 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.2.

3.2 Standard vehicle and accessories. Except as specified herein, the chassis, trailer, components, assemblies, and accessories to be delivered under the contract shall be specified standard or optional items which meet or exceed the requirements of this performance specification. The model furnished shall be not older than the manufacturer’s current model on the date of invitation for bids.

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3.3 Materials. Materials used shall be in accordance with the manufacturer's materials specifications for chassis specified herein. The materials shall be capable of meeting all the operational and environmental requirements specified herein. Recovered materials shall be used to the maximum extent practicable (see 4.6).

3.3.1 Hazardous materials. Asbestos, cadmium, and radioactive material shall not be used in this item. Radioactive material is defined by CFR, Title 10, Parts 30 and 40, and other radioactive material in which the specific activity is greater than 0.002 microcuries per gram or the activity per item equals or exceeds 0.01 microcuries.

3.3.2 Construction. Chassis components, chassis sub-assemblies and assemblies shall be fabricated and assembled into a complete unit in accordance with Army drawings specified below (see 4.6):

Drawing	Model
8358991	M103A3
8358992	M103A4
8358998	M200A1
8736243	M514
8736268	M353
8736315	M390C
8736321	M536
8736322	M537
8736324	M390
8736332	XM545
8736417	XM580
8736829	M492A1
8736996	M569A1
8750054	M116A2
8750240	M116A2E1
8750253	M116A3

3.4 **Operating requirements**.

3.4.1 Performance. Unless otherwise specified herein, chassis shall meet performance requirements of this specification when equipped as specified herein, serviced with standard products (see 6.2), loaded with rated payload and coupled to the tactical prime mover specified in table I. Chassis, serviced and equipped for existing climatic conditions, shall operate as specified without special equipment.

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TABLE I. Payload and speeds.

Chassis	Preferred companion prime mover	Payload (lb)	Miles per hour		
			Highway	Gravel roads	Cross-country
M569A1 1/	M151	500	50	35	15
M116A3	HMMWV	2360	50	35	15
M116A2E1	CUCV	2420	50	35	15
M514	M536	2500	50	-	10
M116A2	1-1/4 ton truck	1965	50	35	15
XM545	M36	2800	50	35	10
M103A3	M35	4090	50	35	10
M103A4	M35	4090	50	35	10
M492A1	M35	4200	50	-	15
M390	M35	4000	50	35	10
M390C	M35	4000	50	35	10
M200A1	M35	5000	50	35	15
M353	M54	7000	50	35	15
M536	M52	6800	50	35	10
M537	M35	5000	50	35	10
XM580	M35	4000	50	35	10

1/ The durability test shall be conducted with the mission payload, however, if such payload is not available, the loads cited herein shall be applied.

3.4.1.1 Bearing seals. When fording or operating in mud, sand, and snow seals shall prevent entrance of foreign matter into the bearings, which are exposed to contamination during these operations. All bearing seals shall restrict the leaking of lubricants from bearings (see 4.7.1).

3.4.2 Electrical system.

3.4.2.1 Electrical circuits. Electrical circuits shall maintain continuity from end to end without evidence of internal or external shorts during all operating conditions (see 4.5).

3.4.2.2 Lights. All lights specified in applicable drawings shall operate throughout all operating conditions (see 4.5).

3.4.2.3 Controls. All electrical, mechanical, and hydraulic controls shall operate without malfunction, throughout all ranges of operation, under all operating conditions (see 4.5).

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3.4.3 Turning and tracking ability. The chassis, when coupled to the prime mover operating in its minimum turning circle, shall follow without cramping and without damage to the towed chassis or prime mover and without interference to the towed chassis or prime mover (see 4.7.2).

3.4.4 Grade and slope operation. The chassis shall operate without slipping or upsetting when ascending or descending longitudinal grades up to 40 percent (%) and side slopes up to 20% (see 4.7.3).

3.4.5 Fording ability. The chassis shall be towable through fresh or salt water to a depth great enough to entirely submerge the chassis for periods up to 15 minutes duration without damage to the chassis or its components. This shall be accomplished without the use of special kits and with a minimum of preparation and servicing before and after fording operations (see 4.7.4).

3.4.6 Landing device. Unless otherwise specified (see 6.2), the chassis shall be equipped with swing-up type landing device. The vertically adjustable swing-up type landing device shall be furnished and mounted on the M103A3, M353, M390, M390C, M200A1, M492, M536, and XM580 chassis (see 4.7 thru 4.9.2). The landing device on all chassis shall have adequate capacity to support the fully loaded vehicle on any grade up to 10 degrees in any direction; and shall withstand without damage the strain imposed upon it when coupling or uncoupling from the towing vehicle. The landing device shall extend and retract and when retracted shall provide ground clearance for cross-country operation (see 4.7.5).

3.4.7 Leveling jacks. When specified (see 6.2), the manually operated leveling jacks shall have capacity to raise and support the fully loaded chassis and shall adjust to permit placing the jack bases on the ground and leveling the chassis on an off-the-road site (see 4.7.5).

3.4.8 Stabilizers. Stabilizers such as step jacks on M200A1 and rear support legs (props) on M103A3 and M116A2, shall stabilize the fully loaded chassis, without raising chassis wheels off the ground. The stabilizers shall exhibit adjustments to permit placing their bases to the ground for stabilizing the chassis on an off-the-road site (see 4.7.6).

3.4.9 Service brakes. When specified (see 6.2), the chassis trailer shall be equipped with service brakes. The combined service brakes of the chassis and towing vehicle shall, under all conditions of loading, bring the combination to a stop within a distance of 30 feet (ft) from a speed of 20 miles per hour (mph) on dry, smooth, level road, free of loose material (see 4.7.7).

3.4.10 Parking brake. The parking (hand) brakes shall be capable of holding the chassis on a dry, hard surfaced, 30% grade with payload headed up or down grade (see 4.7.8).

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3.4.11 Automatic brake actuation. An automatically actuated device shall apply the chassis brakes upon breakaway from the towing vehicle, and shall maintain application of brakes for not less than 15 minutes (see 4.7.8.1).

3.4.12 Payload, towing speeds, and trailing ability.

3.4.12.1 Highway (paved) and gravel roads, payload and speeds. When loaded, the chassis shall be towed over relatively level, smooth, improved, and prepared hard-surfaced roads and gravel roads at average speeds specified in table I, without damage to chassis or prime mover. When loaded and towed at speeds specified, in a straight line, on a level, smooth, paved surface, the path of the towed chassis shall not deviate more than 3 in. to either side of the path of the towing vehicle. When the payloads for trailers M514, XM545, M390, M390C, M536 and M537 are simulated, the loads shall be uniformly distributed to result in 450 to 550 pounds (lb) of static weight over the lunette (see 4.7.9).

3.5 **Support and ownership requirements**.

3.5.1 Safety chains. The trailer shall be equipped with two safety chains and shall not malfunction (see 4.8.1).

3.5.2 Durability.

3.5.2.1 Chassis with rated payload. Except as specified in 3.5.2.2, chassis with rated payload shall possess not less than 0.60 probability of completing a minimum of 6000 miles of operation without experiencing a failure of principal components or subassemblies. This shall include, suspension (including springs, axles and wheels), brake systems, landing device, leveling jacks (when specified), lunette and frame. Operation shall be conducted in accordance with parameters of table II and shall be apportioned as per table II (see 4.8.2).

TABLE II. 6000 mile test, loaded as specified in table I.

Course	Mileage and speeds	Chassis payload
Hard-surfaced roads	2400 miles at varying speeds up to maximum.	Highway
Gravel and dirt roads	1500 miles at varying speeds up to maximum.	Highway
Level cross-country	1380 miles at varying speeds up to maximum.	Cross-country
Hilly cross-country	600 miles at varying speeds up to maximum.	Cross-country
Belgian block	120 miles at speeds applicable	Cross-country

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3.5.2.2 Operation without payload. The 6000 mile durability test shall include 50 miles of gravel and 30 miles of Belgian block without payload, at start and completion of foregoing operations, for a total of 160 miles (see 4.8.2).

3.5.3 Safety standards. No condition shall exist which may create a safety hazard to operating or maintenance personnel (see 4.8.3).

3.5.4 Servicing and adjustments. The trailer shall be adjusted and serviced for each chassis for immediate operational use, including at least the following: adjust braking system, check electrical system, inflate all tires, and completely lubricate the trailer and all running gear with grades of lubricants recommended in table III for the ambient temperature at the delivery point (see 4.5 and 4.6).

TABLE III. Service products specifications.

Product use	Ambient air temperature	
	-65° to 0°F (-54° to -18°C)	-10° to +115°F (-23° to 46°C)
Oil:		
For hydraulic brakes	MIL-B-46176	MIL-B-46176
For shock absorbers	MIL-PRF-6083	MIL-PRF-6083
For general purpose lubrication	Commercial	MIL-PRF-3150
Grease:		
For sealed bearings	MIL-G-23827	MIL-G-23827
For general chassis lubrication, including wheel bearings	MIL-PRF-10924	MIL-PRF-10924

3.5.5 Surface finish. The vehicle component assemblies shall be cleaned, treated, and painted in accordance with the manufacturers standard practices (see 4.5). The paint top coat color shall be as specified in the contract or order (see 6.2). If chemical agent resistant coating (CARC) is required, it shall be specified in the contract or order (see 6.2). When specified (see 6.2), camouflage pattern shall conform to the pattern specified.

3.5.6 Identification and marking. Identification and marking shall be permanent and legible and shall include:

a. Chassis model number
b. Manufacturer’s identification
c. Part or identifying number (PIN)

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Special markings, if required by the applicable military activity, shall be specified in the contract or order (see 4.5 and 6.2).

3.6 **Operating environment requirements**.

3.6.1 Environmental. The chassis shall operate under extreme conditions of weather and altitude in ambient air temperature ranging from minus (-) 50 to plus (+) 125 degrees (°) Fahrenheit (°F). The complete chassis, when in storage, shall withstand a temperature of -80°F without deterioration that may cause failure of any component part of the chassis (see 4.9.1).

3.6.2 Corrosion protection. The vehicle shall be fabricated from compatible materials providing corrosion protection and coating adherence equal to or exceeding that provided by hot dip galvanized 1010 steel, with a minimum coating thickness of 2 mils (50 micrometers) on steel sheet less than 0.0625 inches (in.) (1.6 mm) thick, 2.5 mils (64 micrometers) on steel sheet greater than 0.0625 in. thick, or 0.75 mil (19 micrometers) on pre-galvanized steel sheet 0.0625 in. thick or less (see 4.9.2).

3.6.3 Fungus. Only materials which are fungus non-nutrient shall be used (see 4.6).

3.6.4 Ozone resistance. When rubber components are required to be ozone resistant, the degree of resistance shall be as specified in applicable specifications, or drawings (see 4.6).

3.6.5 Air pollution control. The vehicle shall comply with Environmental Protection Agency (EPA) Regulations governing control of air pollution from new motor vehicles engines in effect on the date of manufacture. In addition, vehicles destined for California shall comply with State of California regulations governing air pollution control in effect on the date of manufacture (see 4.6).

3.6.6 Sound level. The interior sound level shall be applicable to FMCSR (see 4.5).

3.6.7 Rustproofing. When specified (see 6.2), the vehicle shall be rustproofed in accordance with best commercial practices (see 4.5).

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2).
b. Conformance inspections (CI) (see 4.3).

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4.2 First article inspection. First article inspection shall be performed on the first production chassis and two additional chassis randomly selected from the first months production or from the first 20 produced when a first article is required (see 3.1). This inspection shall include the examinations of 4.4, a 5 mile road test (see 4.3.3), and the tests listed within table IV.

TABLE IV. Classification of inspection.

Title	Requirement	Inspection	First article	Conformance tests
Materials	3.3	4.6		X
Construction	3.3.2	4.6		X
Operating requirements	3.4	4.7		
Bearing seals	3.4.1.1	4.7.1	X	X
Electrical system	3.4.2	4.5		
Electrical circuit	3.4.2.1	4.5		X
Light	3.4.2.2	4.5		X
Control	3.4.2.3	4.5		X
Turning and tracking ability	3.4.3	4.7.2	X	X
Grade and slope operation	3.4.4	4.7.3	X	
Fording ability	3.4.5	4.7.4	X	
Landing device	3.4.6	4.7.5	X	X
Leveling jacks	3.4.7	4.7.5	X	X
Stabilizers	3.4.8	4.7.6	X	X
Service brakes	3.4.9	4.7.7	X	X
Parking brakes	3.4.10	4.7.8	X	X
Automatic brake actuation	3.4.11	4.7.8.1	X	X
Highway operation	3.4.12	4.7.9	X	
Support and ownership requirements	3.5	4.8		
Safety chains	3.5.1	4.8.1	X	X
Durability	3.5.2.1 - 3.5.2.2	4.8.2	X	
Safety standards	3.5.3	4.8.3		X
Servicing and adjustments	3.5.4	4.5 & 4.6		X
Surface finish	3.5.5	4.5		X
Identification and marking	3.5.6	4.5		X
Operating environment requirements	3.6	4.9		

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TABLE IV. Classification of inspection - Continued.

Title	Requirement	Inspection	First article	Conformance tests
Environmental	3.6.1	4.9.1	X	
Corrosion protection	3.6.2	4.9.2		
Fungus	3.6.3	4.6		
Ozone resistance	3.6.4	4.6		X
Air pollution control	3.6.5	4.6		
Sound level	3.6.6	4.5		
Rustproofing	3.6.7	4.5		X

4.3 Conformance inspection. Conformance inspection shall include examinations of 4.4, a 5 mile road test (see 4.3.3), and the tests specified in accordance with table IV.

4.3.1 One hundred percent inspection. Unless otherwise specified (see 6.2), each vehicle shall be subjected to the conformance inspection specified in 4.3.

4.3.2 Sampling. Sampling from an inspection lot for conformance inspection shall be specified in the contract (see 6.2).

4.3.3 Road test. Chassis shall be completely assembled, serviced, and operated for a distance of not less than 5 miles, without payload, by the contractor. Performance of chassis shall be demonstrated on relatively smooth, hard-surfaced roads. After completion of 5 miles road test, chassis shall be examined for leaks, brake adjustments, and any other deficiencies. NOTE: If sampling for CI (see 4.3.2) is specified and authorized, road test will not be waived, but will be performed on every chassis by contractor for safety purposes.

4.4 Examination. Each chassis shall be examined for compliance with the requirements specified in 4.7.2 through 4.7.9, and 4.8.2. Any redesign or modification of the contractor's standard product to comply with specified requirements, or any necessary redesign or modification following failure to meet the specified requirements shall receive particular attention for adequacy and suitability. This element of inspection shall encompass all visual examinations and dimensional measurements. Noncompliance with any specified requirement or presence of one or more defects preventing or lessening maximum efficiency shall constitute cause for rejection.

4.5 Verification methods. Acceptable verification methods included in this section are visual inspection and measurement, sample tests, full-scale demonstration tests, simulation, modeling, engineering evaluation, component properties analysis, and similarity to previously-approved or previously-qualified designs.

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4.5.1 Verification alternatives. The manufacturer may propose alternative test methods, techniques, or equipment, including the application of statistical process control, tool control, or cost effective sampling procedures to verify performance. See the contract for alternatives that replace verification methods required by this specification.

4.6 Materials and construction. The materials and construction shall be determined by inspection of contractor records providing proof or certification that materials and design conform to requirements. Applicable records shall include drawings, specifications, design data, receiving inspection records, processing and quality control standards, vendor catalogs and certifications, industry standards, test reports, and rating data.

4.7 **Operating requirements verification**.

4.7.1 Bearing seals. All bearings, as applicable, shall be examined for lubricant leakage and for contamination of lubricant. Any leakage or contamination shall cause for rejection.

4.7.2 Turning and tracking ability test. The prime mover shall be coupled to the chassis and driven to the prime mover's minimum turning circle to the right and to the left, without interference between the prime mover and the towed chassis. During road test, chassis tracking ability shall be observed.

4.7.3 Grade and slope operation. The chassis shall be operated on 40% grades headed up and down grade, and on 20% side slopes with each side of chassis up-slope.

4.7.4 Fording ability. The chassis shall be towed through water crossings at depth and for the time specified. Chassis and components shall subsequently be examined for water damage.

4.7.5 Landing device and leveling jacks. The chassis shall be placed on a firm surface and the landing device and leveling mechanisms operated. The towing vehicle shall be coupled and uncoupled and the leveling mechanism raised and lowered.

4.7.6 Stabilizers test. The chassis, with cross-country payload specified in table II, shall be stabilized as specified on an off-the-road site. Bases shall be extended to the ground to confirm adjustability.

4.7.7 Service brakes. The vehicle combination shall be operated as specified and observed for distance to stop chassis and towing vehicle. Brake stopping distance exceeding the distance specified shall constitute a failure.

4.7.8 Parking brake. The parking (hand) brake shall be actuated with the chassis on a 30% grade, as specified. With service brakes deactivated and trailer coupled to prime mover, set

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trailer parking brakes. Move prime mover to provide clearance between pintle and lunette and set prime mover brakes. Any movement at pintle/lunette couple shall cause for rejection.

4.7.8.1 Automatic brake actuation. The fully loaded trailer, coupled to prime mover, shall be parked on a 15% grade. With prime mover brakes applied, disconnect air lines to trailer, allowing actuator to engage trailer brakes. For inertia brake systems, engage trailer brakes by pulling breakaway chain. The prime mover is to be moved to provide clearance between pintle and lunette so that trailer movement is possible. Reset prime mover brakes; trailer must remain stationary for 15 minutes. Any movement prior to 15 minutes shall constitute test failure. Perform this test both uphill and downhill.

4.7.9 Highway operation. The chassis shall be loaded with the highway rated payload, or with a simulated load of equal weight, uniformly distributed over load area. The chassis shall be operated as specified on paved and gravel roads and observed for towing and performance requirements. Any sign of damage to the chassis or prime mover, or any deviation more than 3 in. to either side of the path of the towing vehicle shall constitute a failure and cause for rejection.

4.8 **Support and ownership requirements verification**.

4.8.1 Safety chains. The two safety chains per trailer shall conform to FMCSR 393.70(d). Any variance to FMCSR 393.70(d) shall cause for rejection.

4.8.2 Durability (chassis with and without payload). The durability requirement will be verified at a 50% confidence level by subjecting two chassis each to the 6000 mile test (utilizing the binomial distribution, no durability failures for two vehicles are allowed to meet this requirement).

4.8.3 Safety standards. The vehicle and all furnished accessories shall comply with all FMVSS and FMCSR in effect on the date of manufacture.

4.9 **Operating environment requirements**.

4.9.1 Environmental operation. When specified (see 6.2), the conformance shall be determined by operating the chassis in temperatures ranging from -50 to 125°F, and storing it at -80°F. The chassis shall evidence no damage as a result of such operations or storage.

4.9.2 Corrosion protection. A material sample shall be capable of meeting or exceeding the corrosion resistance provided by a galvanized sample (as described in 3.6.2) when subjected to the Mandrel Bend Test of ASTM D522, and followed by the Accelerated Corrosion Test of GM 9540P, Method B, 120 cycles.

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5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department or Defense Agency, or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature which may be helpful, but is not mandatory.)

6.1 Intended use. The chassis covered by this specification are military unique because they are intended for non-commercial general non-tactical use by the Government in transporting and dispensing compressed helium gas from a generating or receiving point to the using site. Because of the strict criteria and physical characteristics, the trailer chassis has no commercial application.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.
b. Model of chassis required (see 1.2).
c. Issue of DoDISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.2.1 and 2.3).
d. If first article inspection is required (see 3.1).
e. Physical performance, other than as specified (3.4.1).
f. Landing device, if required (see 3.4.6).
g. Leveling jacks, when required (see 3.4.7).
h. Service brakes, if required (see 3.4.9).
i. Surface finish color and if CARC paint is required (see 3.5.5).
j. If camouflage pattern and type of pattern is required (see 3.5.5).
k. If special marking is required (see 3.5.6).
l. Rustproofing, if required (see 3.6.7).
m. If sampling for conformance inspection is allowed (see 4.3.1).
n. Environmental operation, if required (see 4.9.1).
o. Packaging requirements (see 5.1).

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6.3 First article preproduction. When requiring a first article inspection, contracting documents should provide specific guidance to offerors. This guidance should cover whether the first article is a first article sample, a first production item, or the number of test items. These documents should also include specific instructions regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Pre-solicitation documents should provide Government waiver rights for samples for first article inspection to bidders offering a previously acquired or tested product. Bidders offering such products who wish to rely on such production testing must furnish evidence with the bid that prior Government approval is appropriate for the pending contract.

6.4 Conformance inspection. Affordable conformance inspection with confidence varies depending upon a number of procurement risk factors. Some of these factors include: Contractor past performance, Government schedules and budget, product material and design maturity, manufacturing capital equipment and processes applies, the controlled uniformity of those processes, labor skill and training, and the uniformity of measuring processes and techniques. During the solicitation, contracting documents should indicate those tests desired from table IV and their designated frequency based on a risk assessment for the procurement.

6.5 Pin structure. The PINs to be used for chassis acquired by this specification are created as follows:

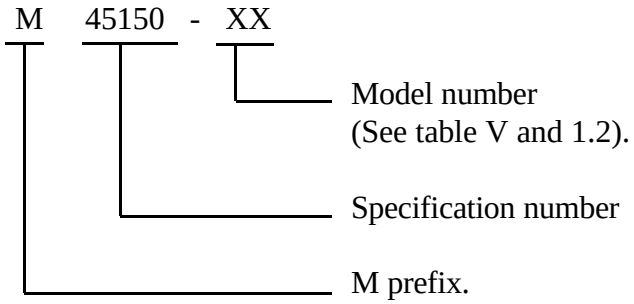


TABLE V. PIN assignment.

PIN number M45150-	Model	PIN number M45150-	Model
01	M569A1	09	M492A1
02	M569A2	10	XM580
03	M116A2E1	11	M390
04	M116A3	12	M390C
05	M514	13	M200A1
06	XM545	14	M537
07	M103A3	15	M353
08	M103A4	16	M536

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6.6 Subject term (key word) listing.

Axles
Bearing seals
Compressed helium
Landing device
Leveling jacks
Non-tactical
Payload
Suspension

6.7 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodians:
Army - AT
Navy - MC

Reviewer Activities:
Army - CR4
Navy – CG

Preparing Activity:
Army - AT

(Project 2330-0317)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE *(YYMMDD)*
	MIL-DTL-45150L	980930

3. DOCUMENT TITLE Chassis, Trailer, 2-Wheel Cart Type 1/4 to 3-1/2 Ton

4. NATURE OF CHANGE ***(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)***

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME *(Last, First, Middle Initial)*	b. ORGANIZATION	
c. ADDRESS *(Include Zip Code)*	d. TELEPHONE *(Include Area Code)* (1) Commercial (2) AUTOVON *(If applicable)*	7. DATE SUBMITTED *(YYMMDD)*

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE *(Include Area Code)*
	(1) Commercial (810) 574-8745 (2) AUTOVON 786-8745
c. ADDRESS *(Include Zip Code)* Commander U.S. Army Tank-automotive and Armaments Command ATTN: AMSTA-TR-E/BLUE Warren, MI 48397-5000	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403 Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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